INCIDENT AND EXPERIMENT REPORT - NO PROCESS CONTACT

The heat came from three headless Python processes - and the experiment they belong to finished on Nov 9

A comprehensive account of the razi-cells e8s1 run, the load still running on the morning of Friday, Nov 13, 2026, and what the files show happened - and did not happen - in between.

Observation window: Monday, Nov 9, 2026, 12:39 p.m. CST to Friday, Nov 13, 2026, 3:55 a.m. CST | Machine: the user's Mac (64 GB memory) | Status at last check: three processes still running, untouched

Summary

The sudden heating was not the open Terminal tabs or Chrome windows. At the time of checking, the Mac's CPU was **0% idle - 88% user and 12% system** - and three headless Python processes were a sustained part of that load. Each was using 50-70% of a core when sampled, each had accumulated about **83 hours and 44 minutes of CPU time**, and the one whose launch was verified, PID 84794, had been started by launchd - not by a terminal session - on Monday, Nov 9 at 12:39 p.m. CST.

That process pointed at the razi-cells experiment folder **e8s1**, specifically its maze cell. The files tell a clean, short story for the experiment itself: **e8s1 finished the same day, Nov 9, with all eight cells complete.** It made 114 model calls, recorded 11 improvements, and reported 923 seconds (15 minutes 23 seconds) of wall time. Seven of its eight cells improved, one of them - fibmod - by a reported 6,709%.

Everything after that, in e8s1, is silence. No file in the e8s1 folder changed after 12:42 p.m. CST on Nov 9, and the temporary folder one process had open no longer exists on disk. In the roughly 87 hours between launch and the Nov 13 check, the three processes together burned about **251 CPU-hours** with no new e8s1 output to show for it.

> **Bottom line:** on the evidence available without touching them, the three processes look **orphaned or stuck spinning**, not productively running. That is an assessment, not a proven diagnosis - the exact loop they are in was not established. Nothing in this report involved interrupting, signalling, sampling again, or otherwise interacting with them after the instruction not to.

Five numbers that carry the report

- **3** Python processes - PIDs 84794, 85007 and 85197 - each single-threaded at observation.
- **87 hours 16 minutes** elapsed for PID 84794, from Nov 9, 12:39 p.m. to Nov 13, 3:55 a.m. CST.
- **About 96%** of one core, continuously: 83 hours 44 minutes of CPU inside that 87-hour window.
- **923 seconds** - the whole useful life of e8s1, against about 251 CPU-hours burned afterwards by the three.
- **0** e8s1 files changed after Nov 9, 12:42 p.m.

1 The CPU had no idle time left, and the Pythons were not the only load on it

The user noticed the machine heating up suddenly, with nothing apparently changed, and ruled out the obvious suspects: the many open Terminal tabs and Chrome windows. The check that followed found a machine with no headroom at all, and a load pattern that did not look like interactive use.

Measure	At observation, Nov 13, early morning CST
CPU overall	**0% idle** - 88% user, 12% system
Three Python processes	Each 50-70% CPU when sampled, 1 thread each, about 83 h 44 min CPU time each
WindowServer	67-124% CPU - the display and window compositor, consistent with a machine under heavy overall strain
A Muse process	About 73% CPU at the same sampling
Memory	52 of 64 GB used, with 14.9 GB of swap in use

Two points matter for reading that snapshot fairly. First, the Pythons were not the whole load: WindowServer and a Muse process were also heavy at that moment, and memory pressure was real. Second, the Pythons were the *persistent* part. Interactive processes rise and fall; these three had each spent close to 84 hours on-CPU, which no bursty foreground task produces.

What "headless" meant here

The user asked whether a headless command-line session was running in the background. For PID 84794, it was: its parent was **launchd**, the system process manager, not a shell in a Terminal tab. Closing tabs would not have stopped it, and it would not have been visible as a window. Its open file at sampling pointed into the experiment's working area:

```
/Users/kethuda/work/razi-cells/exp/e8s1/maze/.tmp/v-x60b5u2l/v2
```

That single path is the thread that ties the heat to a named experiment: razi-cells, run e8s1, cell maze. The other two processes were identified by their near-identical CPU times and contemporaneous sampling, not by individually verified launch records - a distinction this report keeps throughout.

PID	CPU at sampling	CPU time accumulated	Launch verified?
84794	50-70%	about 83 h 44 min	Yes - Nov 9, 12:39 p.m. CST, parent launchd
85007	50-70%	83 h 44 min 08 s	No - inferred same batch
85197	50-70%	83 h 43 min 33 s	No - inferred same batch

The spread between the highest and lowest CPU times is 35 seconds across almost 84 hours. Three unrelated processes do not land that close by accident; a single batch launch, minutes apart at most, is the natural reading. It remains an inference, and it is labelled as one wherever it is used.

2 One batch on Nov 9 was still running 87 hours later

When (CST)	What the record shows
Mon Nov 9, 12:39 p.m.	PID 84794 launched, parent launchd. Pointing at the e8s1 maze cell's temporary working folder.
Mon Nov 9, 12:42 p.m.	Last change to any file in the e8s1 folder. The run's results were complete by this point (see Section 3).
Nov 9-13	No further e8s1 file changes at all, across three and a half days.
Tue Nov 10 - Wed Nov 11	Elsewhere in the razi-cells *exp/* tree, work did move: L1, SP3 and other runs, with results and notes updated through Nov 11. That activity is separate from e8s1 and, on the file evidence, not the output of the three spinning processes.
Fri Nov 13, about 3:53 a.m.	User reports sudden heating. Machine found at 0% CPU idle.
Fri Nov 13, 3:55 a.m.	Reference point for the elapsed-time calculation: 87 hours 16 minutes since PID 84794's launch.
Fri Nov 13, after 3:59 a.m.	On the user's instruction, no further interaction with the processes. Activity Monitor was left open and nothing was killed.

The arithmetic of a stuck core

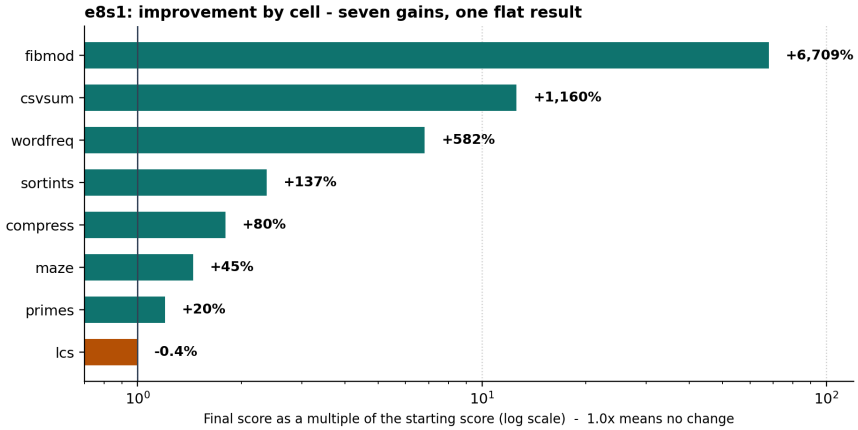
PID 84794 was alive for 87 hours 16 minutes and spent 83 hours 44 minutes of that actually running on a CPU. The ratio is 95.9% - in plain terms, it held roughly one full core, essentially without pause, for three days and fifteen hours. Short sleeps, brief waits, or a process mostly sitting idle waiting for work cannot produce that ratio.

Multiplied across the batch, the three processes account for about 251 CPU-hours (3 x 83.7 hours). That is the heat: not a spike, but a small, constant furnace - roughly three cores' worth of sustained work, day and night, through Nov 10, 11 and 12, on a machine that was also carrying a full interactive workload and swapping heavily.

> **A timing wrinkle, stated rather than smoothed over.** The run reports 923 seconds of wall time, yet its files were last changed three minutes after PID 84794's recorded launch. Those two figures may measure different segments - a run clock inside the experiment versus file timestamps outside it - but the record available for this report does not reconcile them. Both figures are reported as recorded; neither has been adjusted to fit the other.

3 e8s1 finished all eight cells - and fibmod did almost all of the running

Razi-cells, in this run, worked on eight fixed coding tasks - its cells - and tried to improve each one. e8s1 closed with **all eight cells done, 114 model calls made, and 11 improvements recorded**, in a reported 923 seconds. That is about ten model calls per improvement, and a run short enough to sit inside a coffee break - which is precisely what makes the following three days of spinning stand out.



Improvement by cell, converted to a multiple of the starting score (1 + percentage / 100) and plotted on a log scale, because the range - from 0.996x to 68.09x - cannot be read on a linear axis. Labels carry the original reported percentages. Only maze has published starting and finishing scores in the record used here, so no other baselines have been invented for this chart.

Cell	Reported change	Multiple	What is known beyond the number
fibmod	+6,709%	68.09x	Largest gain in the run by an order of magnitude
csvsum	+1,160%	12.60x	Second-largest gain
wordfreq	+582%	6.82x	-
sortints	+137%	2.37x	-
compress	+80%	1.80x	-
maze	+45%	1.45x	Score 48.07 to 69.59 (gain 44.8%, reported as +45%). Winning approach: bidirectional breadth-first search. It cleared 7 small tests plus four 2,000 x 2,000 mazes in about 1.43 seconds in total.
primes	+20%	1.20x	Smallest positive gain
lcs	-0.4%	0.996x	The one cell with no gain - effectively flat

Read as a result, e8s1 is a lopsided success. The headline percentages on fibmod, csvsum and wordfreq suggest tasks where the starting version left enormous performance on the table, while maze - the cell tied to the still-running process - improved by a solid but ordinary 45%, and lcs did not improve at all. A percentage alone does not say how hard a task was or how good the final code is in absolute terms; for seven of the eight cells, the record used here gives no starting or finishing scores, so this report does not supply them.

4 After 12:42 p.m. on Nov 9, e8s1 produced nothing at all

The strongest evidence in this report is negative evidence, and it is unusually clean:

- **No file in the e8s1 folder changed after Nov 9, 12:42 p.m. CST.** Not a result file, not a note, not a log - nothing, across the following three and a half days.
- **The temporary working folder is gone.** The path PID 84794 had open - the maze cell's *.tmp/v-x60b5u2l/v2* - no longer exists on disk. A process can keep working with a file or folder that has been removed from the directory listing, which fits a worker continuing to spin on something the rest of the system has already cleaned up or moved past.
- **The run had already reported completion.** Eight of eight cells done is a finished state, not a halfway checkpoint awaiting three more days of computation.

Put together: whatever the three processes were doing after Nov 9, they were not writing experiment results, and the experiment they were launched for no longer needed them - if it ever needed three of them at all beyond its own fifteen minutes.

The rest of razi-cells did move - elsewhere, and later

The wider programme was not dormant. On Nov 10 and 11, other runs in the same *exp/* tree - named in the record as L1, SP3 and others - produced results and updated notes through Nov 11. That matters in two directions. It shows the research itself continued after e8s1, so "the programme ended on Nov 9" would be wrong. It also sharpens the problem: the later work lives in other runs' folders, while e8s1's folder stayed frozen and its processes kept burning cores. On the file evidence, the Nov 10-11 progress did not come from these three processes.

5 This is the programme recorded in Moissanite-Labs/weft

Yes - e8s1 belongs to the same razi-cells programme documented in Moissanite-Labs/weft, in the record at *docs/program/experiments/razi-cells-laws-2026-11*. e8s1 is one of its local evolutionary-variant runs: variants of code are proposed, tested against fixed tasks, and kept when they improve the score - the pattern visible in miniature in Section 3's eight cells and eleven improvements.

That record is a **run record, not a paper**. It is not checked by continuous integration, and it cannot be reproduced from the folder alone: it depends on the author's own work-directory layout, on Exo imports, and on a live model router. Its stated limits are part of its findings - one model family (Anthropic), and small experimental arms: SP3 ran four runs per arm, with a smallest reported p-value of 0.21, and L1 ran three runs per arm. Small arms like those can show a direction; they cannot settle a law, and the record does not pretend otherwise. Instrument faults and dated corrections are kept inside it - including an invalid E12 maze instrument - which is the right practice, and also a reason to read any single run, e8s1 included, as one observation rather than a verdict.

6 What the wider razi-cells record found

The findings below were verified on Nov 10, 2026 against the weft record itself. They are the programme-level context e8s1 sits inside - not results produced by e8s1, and not results produced by the three processes after Nov 9.

Finding	Numbers behind it
Heredity - passing variants down - cut energy use but bought no extra reach	Energy down 31%; reach 2.27 with heredity against 2.28 without; 65 of 70 heirs changed nothing
Showing cells the model price list changed behaviour immediately	14 of 16 cells chose the cheapest model at the first step when prices were shown, against 0 of 16 when prices were hidden
Hiding the price list was expensive and deadly	Energy cost 2.4 to 3.0 times higher, with 7 cell deaths against 0 when prices were shown
A "stem cell" that fed the others added cost and no reach	Recorded as a negative result in the programme record
Earlier exploratory runs linked variant-sharing to improvement very tightly	13 colonies, 8 fixed coding tasks, about 1,560 model calls, 5 model conditions; variants-share against improvements correlated at r = 0.991 across 8 runs, as recorded in TESTS.md

There is an uncomfortable echo between the second finding and this incident - and it should be stated as an echo, not a cause. A programme that studies how model-driven cells spend energy left three of its own workers spending energy at full rate for three days with no output. Nothing in the record shows the price-list or heredity results caused, predicted, or would have prevented that. The echo is worth noticing; it is not evidence.

7 Orphaned or stuck is the best fit - and it is still an assessment

Evidence for. A completed run (eight of eight cells); a frozen results folder; a vanished temporary folder; a parent in launchd rather than a live terminal session; and a 95.9% on-CPU ratio sustained for 87 hours. Each point alone could have an innocent explanation. Together they describe processes whose supervising work has finished and gone, while the workers themselves never exited.

What else it could be. A retry or polling loop waiting on something that will never arrive - a deleted folder, a finished parent, a model-router response - would look exactly like this from outside. So would a legitimate but silent long computation writing only at the very end, although e8s1's own completed status and frozen folder argue against that here. A bug that loops without output and a job that loops by design are indistinguishable on CPU time alone.

What was not established. The exact code path or loop each process is in; whether all three are in the same state; the launch times and open files of PIDs 85007 and 85197 individually; and whether any of the three has written anywhere outside the e8s1 folder. Answering any of those would have required interacting with the processes or sampling them again, which the user ruled out. This report does not fill those gaps with guesses.

8 Nothing was interrupted - the next move needs your word

For the record, plainly: after the instructions *"dont interrupt without asking"* and *"dont interrupt them or interact with them,"* no process was killed, signalled, paused, re-sampled, or attached to. Activity Monitor was left open on the machine. This report was assembled only from observations and file records already gathered before that instruction, plus the programme record verified earlier in November. The state of the three processes after about 3:59 a.m. CST on Nov 13 is therefore **not claimed** here - they may still be running exactly as found, and this report does not assert a later check that never happened.

Choices, in order of increasing contact - none taken

1. **Leave them running.** Costs roughly three cores and the heat that prompted this report, indefinitely, on current evidence for no output. Preserves whatever state they hold, untouched.
2. **One read-only look, with permission.** A single fresh sample of each process - what code it is executing, what files it now has open - would likely settle "stuck loop" against "working silently," without stopping anything. This is still interaction, which is why it was not done.
3. **Stop the batch.** Ends the heat and reclaims the cores. Also ends any chance of inspecting live state first, and - at a 35-second spread in CPU times - the three should be treated as one batch decision, not three separate ones.

A practical sequence, if you want one: permit the read-only look first, keep its output, then decide on stopping with the diagnosis in hand. Reversing that order destroys the evidence the decision would want.

Appendix - method, provenance and open questions

Method. Process figures, the launch record for PID 84794, the open-file path, and the system snapshot come from the live check made on Nov 13, 2026 before the no-contact instruction. e8s1 figures - eight cells, 114 model calls, 11 improvements, 923 seconds, per-cell percentages, maze 48.07 to 69.59 and its bidirectional breadth-first search detail, the frozen folder after Nov 9, 12:42 p.m., and the missing temporary folder - come from the experiment's own files as read in that same session. Programme findings and limits in Sections 5 and 6 come from the razi-cells record in Moissanite-Labs/weft (*docs/program/experiments/razi-cells-laws-2026-11*), verified on Nov 10, 2026, including its own stated limits. Derived figures - 87 h 16 min elapsed, 95.9% on-CPU ratio, about 251 combined CPU-hours, cell multiples, and the maze gain of 44.8% - were computed from those recorded values for this report and can be re-derived from the numbers printed here.

Open questions. (1) What loop, in code, is each process in? (2) Were PIDs 85007 and 85197 launched with PID 84794, and do they belong to the maze cell or to sibling cells? (3) Why do the run's 923-second wall time and its 12:42 p.m. last file change sit only three minutes after the 12:39 p.m. launch? (4) Did any of the three write anything, anywhere, after Nov 9? (5) What supervises this batch - what was supposed to notice a finished run's workers still running, and clean them up?

Scope limit. This report covers e8s1 and the three processes tied to it. It does not audit the L1, SP3 or other Nov 10-11 runs beyond noting they exist and moved, and it makes no claim about the machine's state after the last observation on Nov 13.